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SPECIAL SECTION

Avoiding a Macabre Future for *Macrhybopsis*—Improving Management and Conservation of Chubs

Genetic structure of the Silver Chub indicates distinctiveness of Lake Erie population

Ahmed A. Elbassiouny[1,2] | **João Pedro Fontenelle**[2,3] | **Patrick M. Kočovský**[4] | **Nicholas E. Mandrak**[2] | **Nathan R. Lovejoy**[1,2]

[1]Department of Cell and Systems Biology, University of Toronto, Toronto, Ontario, Canada

[2]Department of Biological Sciences, University of Toronto Scarborough, Toronto, Ontario, Canada

[3]Institute of Forestry and Conservation, University of Toronto, Toronto, Ontario, Canada

[4]U.S. Geological Survey, Great Lakes Science Center, Huron, Ohio, USA

Correspondence
Ahmed A. Elbassiouny and Nathan R. Lovejoy
Email: ahmed.elbassiouny@utoronto.ca and nathan.lovejoy@utoronto.ca

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Abstract

Objective: Silver Chub *Macrhybopsis storeriana* is a small riverine minnow endemic to North American fresh waters. Its range extends from the southern USA to southcentral Canada; the latter includes a rare lacustrine population in Lake Erie. Anthropogenic activities pose an immediate threat to several Silver Chub populations, currently categorized from special concern to threatened at the state level in the USA and federally and provincially not-at-risk to endangered in Canada. Several studies have examined the anthropogenic causes for the decline of Silver Chub populations, but conservation efforts have been hindered by the lack of knowledge of the population genetics of this species.

Methods: Here, we provide an assessment of the genetic diversity of Silver Chub populations across the USA and Canada using a fast-evolving mitochondrial gene, with particular focus on the Lake Erie population.

Result: We found the Lake Erie population to be divergent from all other populations, with nearly all the haplotypes sampled there being private.

Conclusion: Our study provides genetic evidence that the Silver Chub population in Lake Erie could be considered a separate conservation unit.

KEYWORDS

genetics, molecular phylogenetics, conservation, Silver Chub

INTRODUCTION

The Silver Chub *Macrhybopsis storeriana* is a benthic cypriniform endemic to North America with a maximum total length of 231 mm (Page and Burr 2011). The American range of this species extends from the Missouri River basin in the state of Nebraska in the west to the Monongahela River in the state of Pennsylvania in the east (Kočovský 2019) and from the Red River in the states of Minnesota and North Dakota in the north to the lower Mississippi River in the state of Louisiana in the south (Lee et al. 1980; Werner 2004; COSEWIC 2012). In Canada, disjunct Silver Chub populations occur in the Saskatchewan–Nelson River region, including Lake Winnipeg and the Assiniboine and Red rivers, in the province of Manitoba and in the Great Lakes–Upper St. Lawrence River region, including Lake Erie and occasionally Lake St. Clair and the southernmost tip of Lake Huron in the province of

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Ontario (COSEWIC 2012). These Canadian freshwater biogeographic zones represent the northernmost distributions of this species (Page and Burr 2011).

The Silver Chub population in Lake Erie is particularly interesting as it represents the largest lake-inhabiting population of the species; most other populations occur in rivers (Kočovský 2019), although the species has been occasionally reported from other lakes, likely as vagrants, such as Lake Winnipeg and Lake St. Clair (COSEWIC 2012). Following the Wisconsinan glacial period, Silver Chub likely recolonized Lake Erie <12,000 years ago from a Mississippian refugium, probably through the Michigan basin via stream capture near the Fort Wayne outlet (Bailey and Smith 1981; Mandrak and Crossman 1992). However, no supporting genetic evidence is available for this hypothesis.

Silver Chub was abundant (roughly 8–80 adults per hectare; Kinney 1954) in Lake Erie until the early 1950s, after which the population declined rapidly to roughly less than four adults per hectare (COSEWIC 2012). A prolonged period of lake stratification, caused by prolonged high temperatures and low wind, likely led to the decline in mayflies *Hexagenia* spp. to near extirpation (Britt 1955). As a result, the Silver Chub population presumably declined rapidly as mayflies are known to be the preferred food source (Kinney 1954; Nicholls 1997; Kočovský 2019). More recent studies have further postulated that excess nutrient input, due to pollution, was an underlying factor of water quality deterioration that likely led to the decline in mayflies (Allinger and Reavie 2013). Today, in Canada, Silver Chub has been listed as federally and provincially endangered in Lake Erie in the province of Ontario and not at risk in the province of Manitoba. In the USA, it is listed as extirpated in the state of New York, imperiled or critically imperiled in the states of South Dakota, Nebraska, Michigan, and Georgia, vulnerable in the states of Arkansas, Kansas, Texas, Wisconsin, and West Virginia, but of no special status in the states of Alabama, Illinois, Indiana, Kentucky, Louisiana, Mississippi, Missouri, Ohio, Pennsylvania, and Tennessee (McKenna and Kočovský 2020; NatureServe 2022). Despite government investment in protecting Silver Chub (Boyko and Staton 2010), conservation efforts lack an in-depth understanding of the genetics of Silver Chub populations. Identifying distinct populations, termed evolutionarily significant units (Waples 1995; Waples et al. 2001), is crucial for protecting biodiversity below species level and prioritizing populations in immediate need of conservation and recovery (Mace et al. 2007). In Canada, such biodiversity is legally termed "designatable units" and must fulfill discreteness and evolutionary significance criteria (COSEWIC 2021). The Silver Chub population in Lake Erie meets both designatable unit criteria: it is isolated from all other populations (discreteness) and it is the only established lacustrine population, meeting the evolutionary significance criteria (COSEWIC 2012). However, genetic information supporting the discreteness of the Lake Erie population is still lacking.

Impact statement

Lake Erie supports more than 11 million residents and ~ USD $50 billion industries, but pollution and associated algal blooms have damaged this ecosystem and put many species, like Silver Chub, at risk. This study determined that Lake Erie Silver Chub is genetically distinct from all other populations, which will help guide conservation efforts in restoring this natural resource.

Identifying and characterizing genetic diversity is crucial for conservation practices (Johri et al. 2022). Identifying genetic diversity within and among populations can provide insights on population connectivity and substructure (Kimura 1986; Slatkin 1987). Connectivity between populations can be inferred based on genetic differences between populations using *F*-statistics (Weir and Cockerham 1984), allowing for inferences of genetic variability and, consequently, population viability and adaptive potential (Barrett and Schluter 2008; Kardos and Luikart 2021). Similarly, haplotype diversity (h) (Nei 1987) and nucleotide diversity (π) (Nei and Li 1979) provide information on population genetic diversity: the former indicates the probability of two randomly chosen haplotypes being different (Nei 1987), and the latter indicates the average number of nucleotide differences per site between two randomly chosen sequences (Nei and Li 1979).

To determine if the Silver Chub population in Lake Erie warrants protection as a designatable unit, we examined the genetic structure of Silver Chub across its range. We used a fast-evolving mitochondrial gene, cytochrome b, to evaluate phylogenetic structure and haplotype variation and to determine population differentiation.

METHODS

Fish collection and sampling

Muscle tissues or fin clips were obtained from 134 specimens of Silver Chub collected between 2012 and 2021 (Figure 1; Tables 1 and Supplemental Table 1 [available in the online version of this article]). Lake Erie samples were collected in trawl nets by the U.S. Geological Survey and Ontario Ministry of

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FIGURE 1 Localities of Silver Chub specimens analyzed in this study. Dots represent collection sites colored by the population sampled. The geographic range of Silver Chub (Boyko and Staton 2010) is shown in dark gray.

TABLE 1 Number of Silver Chub samples from different localities sequenced in this study and the representative GPS coordinates of their sampling locations.

Locality	Sample size	Latitude	Longitude
Assiniboine River	22	49.8674	−97.4375
Lake Erie	25	41.8852	−82.8812
Lower Mississippi River	2	31.0271	−91.5785
Upper Mississippi River	24	44.5836	−92.5102
Missouri River	26	40.3649	−95.6392
Ohio River	16	38.7953	−84.1742
Wabash River	19	38.9348	−87.5163

Natural Resources and Forestry in Canadian waters or U.S. Geological Survey and Ohio Department of Natural Resources in American waters in Lake Erie. Specimens were preserved in 70–100% nondenatured ethanol and stored at −20°C. Samples from other localities were collected in trawl nets (lower Mississippi, upper Mississippi, and Missouri rivers), in trap nets (upper Mississippi River), by seine (Wabash River), or by electrofishing (Ohio and upper Mississippi rivers).

Extraction, amplification, and sequencing of DNA

Total DNA was extracted and purified from at least 5 mg of tissues using DNeasy Blood and Tissue extraction kit (Qiagen) following the manufacturer's protocol. A pilot study was first conducted comparing two mitochondrial markers commonly used in phylogenetic studies, cytochrome c oxidase I and cytochrome b; the latter showed higher sequence variation among Silver Chub populations (Supplemental Table 2). A fragment of the mitochondrial gene cytochrome b was amplified from the purified DNA using ExoCB.F and ExoCB.R primers (Lewallen et al. 2011) in a 25.0-μL reaction containing 1× Taq buffer, 4 μM of each primer, 1.5 mM $MgCl_2$, 100 nM each dNTP, and 5 U Taq DNA polymerase (Invitrogen). Amplification followed a 30-s denaturation step at 95°C, followed by 45 cycles of denaturation at 94°C for 30 s, annealing at 54°C for 60 s, and elongation at 72°C for 90 s, and concluded with a final elongation step at 72°C for 5 min.

Successful amplification was confirmed using 1.0% agarose gel electrophoresis. The PCR product was cleaned using an enzymatic reaction of equal amounts of Exonuclease I and Calf-Intestinal Alkaline phosphatase incubated at 37°C for 30 min, followed by an inactivation step at 85°C

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for 15 min. Bidirectional sanger sequencing was completed at the Centre for Applied Genomics at the Hospital for Sick Children (Toronto, Ontario) on the purified PCR product.

Sequence analyses

Sequence reads were checked for quality and trimmed in Geneious Prime (www.geneious.com). Sequences were subsequently aligned using a multiple sequence alignment method (program MUSCLE; Edgar 2004). Three individuals had poor-quality sequences and were excluded from the haplotype and genetic distance analyses described below (but were included in the phylogeny). We added an *M. storeriana* sequence from Genbank (accession number: NC_030485.1), as well as a sequence for the closely related species *M. hyostoma* as an outgroup (accession number: NC_030610.1). Thus, our total data set for haplotype analyses included 132 new sequences but 136 for the phylogenetic inference. Phylogenetic analysis was performed using a maximum likelihood algorithm (IQ-Tree; Nguyen et al. 2015) with 1000 replicates of SH-like approximate likelihood ratio test (Guindon et al. 2010) and 1000 bootstrap replicates. We used IQ-Tree to test the fitness of substitution models to our data set, and the best-fit model, K3Pu + F + R2, was used in the analysis.

We used the packages *haplotypes* version 1.1.2 (Aktas 2020) and *pegas* version 1.1 (Paradis 2010; Paradis et al. 2021) in R to determine haplotypes and construct haplotype networks, respectively. Haplotype and nucleotide diversities were estimated using *pegas* in R. Average p-distances, the proportion of sites in an alignment where two sequences differ, averaged within and between populations were calculated using a molecular evolutionary genetics analysis (MEGA; Tamura et al. 2021). To infer population differentiation, we estimated pairwise F_{ST} values between populations following Weir and Cockerham (Weir and Cockerham 1984) using the R package *hierfstat* (Weir and Goudet 2017; Goudet et al. 2021). We tested for significance of calculated pairwise F_{ST} values between populations using *G*-statistic permutations (function *test.g*) and bootstrapping permutation over loci of pairwise F_{ST} (function *boot.ppfst*) in *hierfstat*.

RESULTS

Silver Chub mitochondrial haplotypes

Our haplotype data set includes 132 Silver Chub sequences of 746 bp length from locations spanning six tributaries from the lower Mississippi River to the Assiniboine River (Table 1; Figure 1). Sequences have been deposited in Genbank, with accession numbers OQ215578–OQ215711 (see Supplemental Table 1). The sequences represented 36 haplotypes (Figure 2 and Supplemental Figure 3 [available in the online version of this article]). Several haplotypes were shared among localities, including two of the most common haplotypes (VII and XIV), observed respectively in 13 and 39 individuals. Haplotype VII was observed in the Assiniboine, Wabash, and upper Mississippi rivers, and haplotype XIV was distributed across the Assiniboine, upper Mississippi, Missouri, Ohio, and Wabash localities. Haplotype X was also observed in high frequency (15 individuals) but was only found at the Lake Erie locality.

The haplotype network (Figure 2) shows that samples from Lake Erie are genetically distinct. Lake Erie has seven haplotypes (including the second most frequent, haplotype X) that were not observed in any other localities and clustered together in the network. In contrast, other localities generally included haplotypes that occurred at multiple localities. A single individual from Lake Erie had a haplotype (XVIII) that was also observed in the Wabash and upper Mississippi rivers (Figure 2). We also note that the Lake Erie cluster of haplotypes is connected to a haplotype that occurs in the Ohio and Wabash rivers (haplotype III).

The maximum likelihood phylogeny (Figure 3) supports a close phylogenetic relationship among all Silver Chub populations and highlights five monophyletic clades with higher bootstrap confidence: (1) Lake Erie clade consisting of all but one sample from Lake Erie; (2) clade comprising one Lake Erie individual (sample 12376) and samples from the upper Mississippi, Ohio, and Wabash rivers; (3) clade consisting of Assiniboine, Missouri, and Ohio River samples; (4) clade consisting of lower Mississippi, upper Mississippi, Wabash, and Ohio River samples; and (5) clade consisting of Missouri, Ohio, and Wabash River samples (Figure 3).

Population variation and divergence

Genetic p-distances within populations ranged between 0.00199 (Lake Erie) and 0.00754 (lower Mississippi River) (Figure 4A). Pairwise F_{ST} values between localities ranged from 0.00 to 0.06 (Figure 4B). The Lake Erie population is the most genetically differentiated from other populations as its pairwise F_{ST} values ranged from 0.04 to 0.06, while the next highest pairwise F_{ST} divergence observed is 0.03 between the Assiniboine River and lower Mississippi River populations. Our *G*-statistics test corroborates that the observed F_{ST} differences between these populations are unlikely to be due to random assortment of individuals but possibly due to genetic structuring (p-value = 0.01). Additionally, the bootstrapping over loci of pairwise F_{ST} test indicates 95% confidence intervals not overlapping zero (confidence intervals

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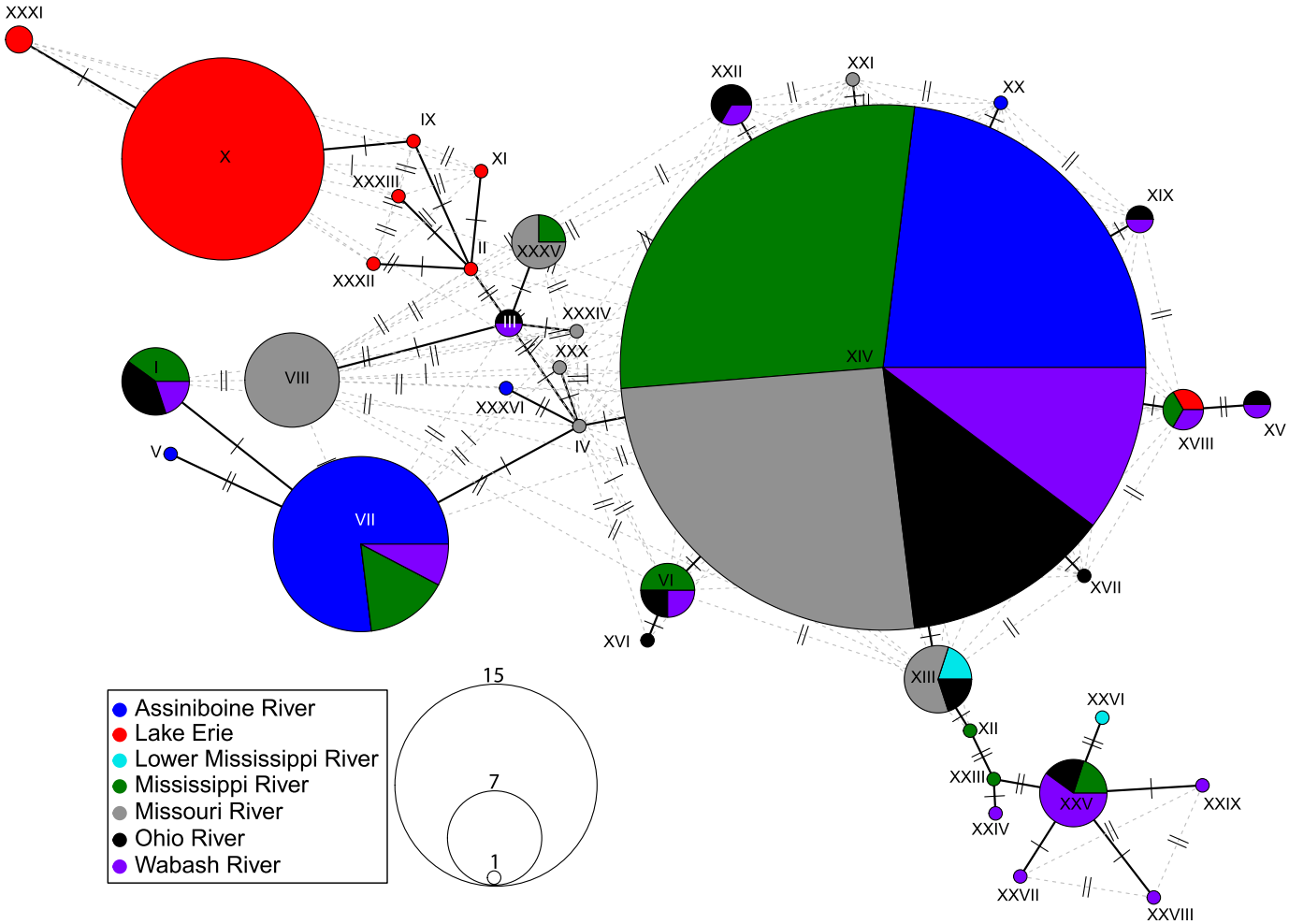

FIGURE 2 Haplotype network for all Silver Chub cytochrome b sequences in this study. Colors for each haplotype represent the localities where it was observed (see Figure 1 for map of collection sites). Hash marks indicate mutational steps between haplotypes. Lines (links) represent the relationship between haplotypes. Full lines indicate links with the least number of mutational changes between haplotypes, while dashed lines indicate alternative links with a threshold of two possible mutations. The size of the haplotype indicates the frequency of the haplotype in our data set. Roman numerals refer to the designated haplotypes in our analyses (see Supplemental Table 1).

do not include $F_{ST} = 0$, which would mean no genetic differences between populations) for every pairwise F_{ST}, with the exception of four pairwise comparisons: upper Mississippi versus Ohio, upper Mississippi versus lower Mississippi, Ohio versus lower Mississippi, and lower Mississippi versus Wabash (Figure 4B), suggesting some degree of population differentiation among all but these four pairs.

Comparison of pairwise p-distances between localities indicate that p-distances between Lake Erie and other locations generally have the highest values (Supplemental Figure 1). The smallest p-distances are among the Ohio, upper Mississippi, Missouri, and Assiniboine populations (Supplemental Figure 1). The haplotype and nucleotide diversity across all localities are 0.885 and 0.004, respectively (Supplemental Table 3). Lake Erie has the highest number of private haplotypes (seven out of eight) but the lowest haplotype diversity (Supplemental Table 3). The Wabash River has the highest haplotype diversity but only 4 out of 14 haplotypes are private.

DISCUSSION

In this study, we provide the first comprehensive analysis of genetic diversity of Silver Chub across its geographic range. We investigated its genetic structure from localities across three major North American drainages: St. Lawrence (Lake Erie draining to Atlantic Ocean), where it is federally and provincially endangered in Canada; Red River (Assiniboine River draining to Lake Winnipeg); and Mississippi River (draining to Gulf of Mexico). We found that Silver Chub collected from Lake Erie, a lake-dwelling population, are genetically distinct from all other populations, suggesting that the Lake Erie population could be treated as a separate conservation unit. Our analyses show that Lake Erie samples have endemic haplotypes not found elsewhere, indicative of low gene flow between the Lake Erie population and other populations. We also found that genetic diversity within Lake Erie is lower than in

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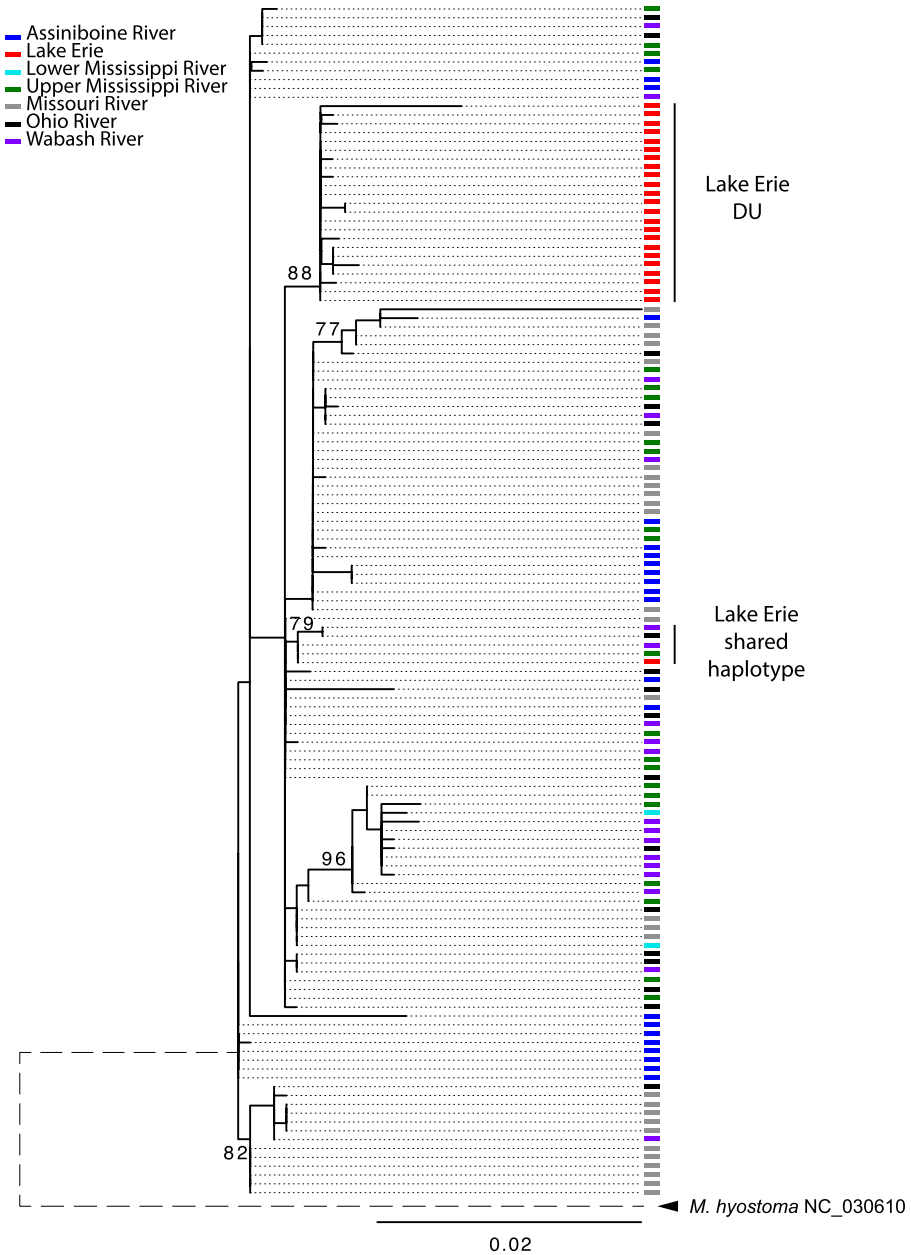

FIGURE 3 Maximum likelihood phylogeny of Silver Chub cytochrome b sequences in this study. Lake Erie samples form a distinct clade. The clade labeled "Lake Erie shared haplotype" includes one sample from the Lake Erie population with shared *cytb* haplotype with specimens from the upper Mississippi and Wabash rivers. The dashed line with a black arrowhead shows the placement of the outgroup (*M. hyostoma* sequence from Genbank), which is not to scale. The five monophyletic clades with bootstrap support above 70% are labeled with support values. See Supplemental Figure 2 for phylogeny with full sequence names and locality information.

other populations, highlighting the vulnerability of this population to extirpation.

Lake Erie population is genetically divergent

We found multiple lines of evidence supporting the distinctiveness of the Silver Chub population in Lake Erie. First, the mitochondrial *cytb* sequences from the Lake Erie population form seven haplotypes endemic to Lake Erie, one of which (haplotype X) was detected in 15 specimens. Second, the genetic distance and F_{ST} scores of Lake Erie compared with other populations were higher than comparisons between other population pairs. Finally, we detected substitutions in *cytb* (sites 442 and 589; Supplemental Figure 3) that are unique to Lake Erie.

Previous biogeographic analysis suggests postglacial dispersal of Silver Chub from a Mississippian refugium into Lake Erie, likely through stream capture

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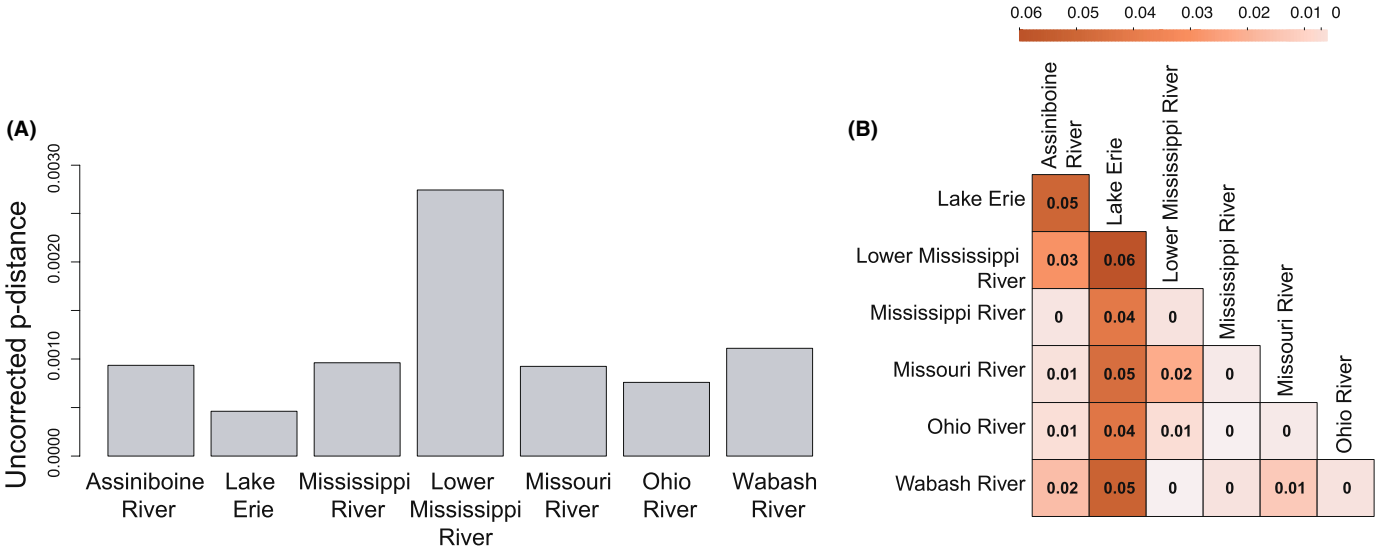

FIGURE 4 Genetic distances and population differentiation, based on cytochrome b sequences, within and among populations of Silver Chub, showing (A) within-population mean uncorrected p-distances among individuals for the seven localities sampled in this study and (B) pairwise F_{ST} values between populations, shaded according to F_{ST} value.

in the vicinity of the Fort Wayne outlet (Mandrak and Crossman 1992). The higher genetic differentiation and lack of shared haplotypes between Lake Erie and the other populations (Figure 4 and Supplemental Figure 1) suggests that contemporary gene flow is not occurring between populations. However, we did detect a single Lake Erie haplotype that was also present in the Wabash and upper Mississippi rivers. This haplotype might represent a remnant of the original diversity associated with the postglacial colonization of Lake Erie. Alternatively, it might represent a secondary, more recent, dispersal into Lake Erie. In the latter case, this haplotype may have originated from the Wabash River via the Maumee River; the Maumee River is known to undergo headwater capture with the Wabash River in the vicinity of Eagle Marsh (USACE 2014). The Maumee River has also been suggested as a possible spawning location for the Lake Erie Silver Chub population (Kočovský 2019). More sampling and genetic analyses of specimens from the Maumee River and adjacent Wabash River would further address these hypotheses. Similar patterns were identified in Eastern Sand Darter *Ammocrypta pellucida*, another endangered freshwater species in the provinces of Ontario and Quebec and the states of New York, Michigan, Pennsylvania, and Vermont, that colonized the Laurentian Great Lakes around the same time as Silver Chub (Mandrak and Crossman 1992). Walter and colleagues found that an Eastern Sand Darter population in West Lake (Ontario) was genetically divergent from those of other populations but similar to Wabash River populations (Walter et al. 2022), suggesting a possible shared dispersal route with Silver Chub.

Lake Erie Silver Chub population is a separate conservation unit

Our data indicate that the Lake Erie population is genetically distinct from other Silver Chub populations in Canada and the USA. Moreover, the levels of genetic differentiation observed, including Lake Erie–specific haplotypes and Lake Erie–specific substitutions, suggest that the Lake Erie population may have been evolving independently for a considerable span of time. Over this time, Lake Erie Silver Chub may have accumulated genetic changes related to lake adaptation. This adaptation to lake dwelling could not be readily restored if the Lake Erie population became extirpated. Additional genetic markers and measurements of adaptive morphological characters (e.g. body shape) are required to further test this hypothesis (Ferguson 2002). This interpretation would support the Committee on the Status of Endangered Wildlife in Canada discreteness criterion (COSEWIC 2021). However, analyses based on neutral markers, such as ours, are not sufficient for assessing evolutionary significance (e.g., local adaptation). Additional analyses based on functional genes could be conducted to confirm the evolutionary significance of the Lake Erie population. In the meantime, the Lake Erie population is the largest lacustrine population of this species, while most other populations exhibit specialized riverine reproductive traits (McKenna and Kočovský 2020; McKenna et al. 2023, this special section), which is indicative of local adaptation.

High haplotype diversity (h = 0.640–1.000), in combination with low nucleotide diversity (π = 0.002–0.009), was found in the riverine populations, suggesting population

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growth following a period of low effective population size (Grant and Bowen 1998). Conversely, high haplotype diversity in the Lake Erie population ($h = 0.581$; $\pi = 0.001$) suggests a stable population, isolated from the riverine ones. Haplotype diversity (h) is sensitive to sample size (Sun et al. 2004), and additional sampling in regions with low sample sizes (for example, lower Mississippi River) could provide a better estimate of genetic diversity and growth.

Genetic diversity of Silver Chub could be affected by habitat fragmentation

We found high haplotype diversity across Silver Chub populations, most of which (67%) are private haplotypes, suggesting limited secondary contacts between some populations. While the Lake Erie population appears to be genetically distinct, we found lower F_{ST} values among other Silver Chub populations despite the significant geographic distances involved. However, the genetic marker we used, and the spatial scale of our sampling, limits our ability to identify potential barriers to recent dispersal and gene flow. Several localities in our study system are separated by human-manufactured barriers. For example, the lower Mississippi River is separated from the upper Mississippi River by a series of locks and dams. Similarly, the Ohio River is interrupted by 19 locks and dams along its length. Higher resolution genetic markers (microsatellites or genomewide SNPs) combined with increased geographic sampling could allow testing of the effects of these structures on current patterns of gene flow in Silver Chub. In addition, study of the effects of barriers on population viability could be beneficial. Silver Chub is thought to broadcast spawn in open waters (Werner 2004; McKenna et al. 2023), and migration is likely an important part of its life cycle (for more details, see Perkin et al. 2023, this special section). Stream fragmentation can affect the abundance, distribution, and community structure of fish species (Perkin and Gido 2011, 2012; Perkin et al. 2015), and riverine species like the Silver Chub may be particularly susceptible.

Although our study clearly demonstrates the genetic distinctiveness of the Silver Chub population in Lake Erie, additional loci, especially from the nuclear genome, and morphological data are needed to confirm the evolutionary distinctiveness of the Lake Erie Silver Chub. Comprehensive genomic sampling using next-generation sequencing would allow evaluation of genetic structure within the Lake Erie population, admixture trends, and population dynamics. Our study is the first to investigate the population genetic structure of Silver Chub, an endangered species in Canada and in many jurisdictions in the USA. Our results suggest that the Lake Erie population of Silver Chub could be treated as a separate designatable unit. We also found heterogeneity in the genetic diversity across populations, and we hypothesize that anthropogenic barriers, such as dams, could be a driver of such disparity.

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CONFLICT OF INTEREST STATEMENT

There is no conflict of interest declared in this article.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on National Center for Biotechnology Information's Genbank, with accession numbers OQ215578 to OQ215711.

ETHICS STATEMENT

There are no ethical guidelines that were applicable to this study.

ORCID

Ahmed A. Elbassiouny https://orcid.org/0000-0003-2372-4928

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